INSIDE THE GAME

Synthetic Football Data & Schema Specification

Hackathon prototype + scalable future architecture | Version 1.0 | 25 October 2026

Concept: An AI-powered Premier League-inspired football intelligence experience that transforms synthetic event streams into explainable player ratings, live tactical narratives, and evidence-based substitution advice.

Scope: Prototype design specification. All sample records and numerical parameters are illustrative, not Premier League data, Opta data, Sofascore weights, or a validated forecasting model.

Recommended approach: Compute football facts deterministically; allow an AI agent to select tools, interpret evidence, explain uncertainty, and tailor communication to fans and analysts.

At a glance

Prototype (build first)	Future expansion
Synthetic match replay; event ingestion; player rating; historical comparison; momentum triggers; bench recommendation; fan/analyst narrative; audit trail	20-club season-scale simulation; richer tracking; calibrated xG/xA/xT; tactical counterfactuals; confidence intervals; broadcaster surfaces; personalised notifications; automated evaluation

Document map: 1. Design principles 2. Scope and features 3. Required synthetic data 4. Relational schema 5. Event model 6. Derived features and ratings 7. Substitution decisioning 8. Agent/workflow 9. Data generation and validation 10. Implementation roadmap 11. Sample records and acceptance criteria

1 | Design principles and important distinctions

• Synthetic events are the source of truth. Goals, passes, duels and turnovers should arise from coherent simulated possessions; do not invent independent summary totals that contradict events.

• Historical baseline != live rating. Historical matches establish form and expected performance by role and minutes; current match events determine the live rating.

• Low rating != automatic substitution. Evaluate tactical need, match state, available bench, opponent matchups, fatigue, risk and uncertainty before proposing a change.

• Explainability is mandatory. Each claim points to event IDs, derived metric IDs, baseline windows and model version. No unsupported causal claims.

• AI does not invent arithmetic. Statistical and football rule calculations live in tested services. The agent retrieves them and narrates findings.

• Never represent synthetic records as real match facts. Label match, player and recommendations as simulated; use fictitious player IDs/names for default demos.

Reference workflow

Synthetic data generator -> fixture, squads and possession/event stream -> event validator -> feature aggregator -> position-aware rating service -> insight triggers and candidate ranking -> AI analyst agent -> human-readable fan/analyst dashboard + traceable evidence.

2 | Product scope: prototype versus expansion

Capability	MVP deliverable	Later enhancement
Event processing	Replay one match with event timestamps; live cumulative stats	Streaming ingestion, replay controls, varied match scenarios
Player ratings	Role-aware, transparent 0-10 illustrative model	Fitted and calibrated against independent simulated outcomes
Historical context	10 prior matches per team, last-5/last-10 baselines	Longitudinal form, seasons, opponent-adjusted strength
Live intelligence	Pressure / momentum / tactical shift alerts	Advanced sequence, contextual rarity, role-specific insights
Substitution adviser	Compare 2-4 eligible bench options with pros/cons	Counterfactual rollout, outcome distributions, uncertainty
AI narratives	Agent-generated explanations with evidence IDs	Multilingual, broadcaster and personal preference formats
Fan & analyst views	Concise story feed + detailed metrics panel	Tailored feeds, audio, visual overlays, user profiles
Movement intelligence	No unsupported tracking-derived claims	Full tracking / positions, off-ball and spacing metrics

3 | Synthetic datasets needed

Dataset	MVP target	Required inputs / rationale
Teams & fixtures	4 fictional clubs; 20 historical fixtures + 1 demo	Team identity, match venue, date, kickoff, opponent, final score
Players & squads	25 players/team (~100)	Role, secondary roles, foot, archetype, ability priors
Historical lineups	Lineups for each historical fixture	Starter/bench, formation, minutes, substitutions, eligibility
Historical events	~30k-40k event records total	Linked sequences with actor, target, defending opponent, timestamps, locations
Live demo events	~1.5k-2k event records	Sorted event replay; possessors, actions, outcomes, key moments
Tactics & game states	Pre-match + changes	Formation, pressing, block, width, score differential, game state
Bench status	Per fixture	Injury/fitness availability, role coverage, substitution windows used
Rating & insight artifacts	Produced by system	Derived metrics, performance deviations, triggers and evidence
Recommendation artifacts	Produced by system	Candidate score, reasons, tradeoffs, uncertainty, alternative/no-change

Scale note: 20 distinct fixtures across four teams imply 10 fixtures per team if evenly distributed. Event counts are illustrative volume targets, not claims about exact real match-event frequencies.

4 | Data architecture and keys

Use PostgreSQL or SQLite for MVP structured data; append-only events in event table and JSON metadata for variable event attributes. Store snapshots, metrics, explanations and model versions separately to enable reproducibility. FK = foreign key; PK = primary key; nullable fields are marked ?.

Core relational schema

Table (PK)	Key fields (suggested types)	Relationships / purpose
teams (team_id)	name TEXT; code TEXT UNIQUE; home_venue TEXT?; strength_index NUMERIC	Referenced by matches, players, lineups
players (player_id)	team_id FK; display_name TEXT; primary_position TEXT; secondary_positions JSON; foot TEXT; archetype TEXT	One player belongs to one team in MVP
matches (match_id)	home_team_id FK; away_team_id FK; kickoff_at TIMESTAMP; status ENUM; home_score INT; away_score INT; synthetic BOOL	Unique fixture and match result
match_rosters (match_id,player_id)	team_id FK; roster_status ENUM; shirt_number INT?; available BOOL; reason TEXT?	Define starter, bench and unavailable
match_player_stints (stint_id)	match_id FK; player_id FK; start_sec INT; end_sec INT?; position TEXT; formation_slot TEXT?	Minutes and position changes; do not double-count
tactical_states (tactic_id)	match_id FK; team_id FK; start_sec INT; end_sec INT?; formation TEXT; press_style TEXT; line_height TEXT	Time-bounded tactical context
possessions (possession_id)	match_id FK; team_id FK; start_sec INT; end_sec INT?; start_event_id?; end_reason TEXT?	Chain events with a controlling team
events (event_id)	match_id FK; sequence_no INT; event_sec INT; period INT; possession_id FK?; team_id FK; player_id FK?; type TEXT; outcome TEXT; x NUMERIC?; y NUMERIC?; recipient_id FK?; defender_id FK?; attrs JSON	Append-only source events; indexed by match/time
player_match_metrics (match_id,player_id,metric_code,asof_sec)	value NUMERIC; unit TEXT; algorithm_version TEXT; source_event_hash TEXT	Derived aggregations, not independent truth
ratings (rating_id)	match_id FK; player_id FK; asof_sec INT; live_rating NUMERIC; expected_rating NUMERIC?; role TEXT; methodology_version TEXT	0-10 rating snapshots with context
insights (insight_id)	match_id FK; asof_sec INT; kind TEXT; summary TEXT; audience TEXT; evidence_ids JSON; agent_version TEXT	Store traceable narratives
recommendations (rec_id)	match_id FK; asof_sec INT; outgoing_id FK?; incoming_id FK?; option_rank INT; confidence TEXT; upsides JSON; downsides JSON; evidence_ids JSON; model_version TEXT	Include no-substitution alternative

Future tables

tracking_frames(frame_id, match_id, time_ms, player_id, x, y, speed, ball_x, ball_y); training_loads(player_id, measured_at, load, availability); player_role_priors(player_id, role, model_version, coefficients); possession_value_models(model_id, version, calibration); scenario_runs(run_id, match_id, candidate_id, seed, outputs); users/user_preferences/notifications; competitions/seasons/contracts; evaluation_cases and labels.

5 | Event schema: minimum canonical contract

Field	Type	Rule / use
event_id	UUID / TEXT	Unique immutable identifier
match_id	FK TEXT	Must reference an existing simulated match
sequence_no	INTEGER	Unique increasing sequence inside match
period	INTEGER	1/2; use extra-time periods in future
event_sec	INTEGER	Second within match timeline; monotonic with sequence
possession_id	FK TEXT nullable	Links football sequence; restart/whistle events may be null
team_id	FK TEXT	Actor team, consistent with lineup
player_id	FK TEXT nullable	Actor when applicable; active at event time
event_type	ENUM/TEXT	pass, carry, take_on, duel, tackle, interception, shot, foul, card, substitution, restart, pressure, recovery
outcome	ENUM/TEXT	success, fail, blocked, saved, goal, out, neutral etc; validated per event type
start_x/start_y/end_x/end_y	NUMERIC nullable	0-100 coordinate system; document pitch orientation
recipient_id / defender_id	FK TEXT nullable	Contextual target or matched opponent
body_part / shot_type	TEXT nullable	Present only for relevant event families
under_pressure	BOOLEAN nullable	Only if simulator explicitly models pressure
qualifiers	JSON	Versioned event-specific detail; avoid untyped sprawl
source_type	ENUM	synthetic_simulation; never mislabel as official
generator_version / seed	TEXT / INTEGER	Reproducibility, provenance

Event taxonomy and conditional attributes

Family	Example types	Typical conditional attributes
On-ball progression	pass, carry, take_on	target player; start/end x,y; pass length; contested; line_break flag derived
Chance creation	key_pass, shot	shot xG feature inputs; assister event ref; shot target; goal linkage
Defensive	tackle, duel, interception, clearance, block	opponent ID; duel type; success; regain possession?
Discipline	foul, card, penalty	card color; offence category; penalty awarded event ref
Keeper	save, claim, punch, distribution	shot ID; location; xGOT inputs where present
Tactical/administrative	substitution, position_change, tactic_change	out/in player IDs; before/after formation; effective timestamp

Modelling rule: Avoid separate independent “assist”, “key pass”, and “goal” events that cannot be reconciled. Derive assist and key-pass tags from linked shot/pass sequences with stable definitions; likewise a blocked shot is counted once with appropriate fields.

6 | Features, ratings and historical form

Derived metrics

Metric	Derived from	Required context
Pass accuracy	completed passes / attempted passes	Exclude invalid/non-attempt actions consistently
Take-on success	successful take-ons / attempts	Defender matchups for matchup analysis
Progressive passes/carries	start and end coordinates and orientation	Choose and version progression threshold
xG / xA	Shot quality model / linked chance-pass model	Shot geometry/context; label as illustrative until calibrated
xT	Location value before vs after action	Versioned grid/state-value model
Pressure regains	pressure event followed by team possession regain	Temporal window + explicit attribution rule
Live rating	Weighted positive/negative event contributions by role	Min-play time, participation opportunity and caps
Historical expected rating	Recency-weighted prior matches conditioned on role/minutes/opponent	Do not leak current/future match events into baseline

Rating approach: Start at an illustrative base (e.g. 6.0), sum role-specific, bounded event contributions, apply context and minutes normalization where appropriate, then clip to [0,10]. This is a prototype formula, not the Opta or Sofascore algorithm. Define positive, negative and overlapping event rules in a versioned scoring configuration.

Form deviation: current live rating - comparable historical expected rating. Display baseline sample size and uncertainty. Warn when a player has played too few minutes or few relevant actions to justify a confident conclusion.

7 | Substitution decision schema and logic

Eligibility and candidate evaluation

Step 1: Validate substitutions remaining, permitted windows, candidate roster status and role availability. Step 2: Diagnose problem from event evidence. Step 3: Create no-change + eligible substitute options. Step 4: Score each option with bounded metrics. Step 5: Explain advantages, disadvantages, assumptions and confidence. Step 6: Recompute after every material event.

Decision dimension	Evidence	MVP rule
Live performance	Rating deviation; failures; trend; minutes	Trigger investigation, never auto-sub
Tactical role fit	Candidate primary/secondary positions; formation	Reject obviously infeasible options

Decision dimension	Evidence	MVP rule
Opponent matchup	Historical/simulated duel styles; location/context	Use relevant role-specific rates, with sample sizes
Game situation	Score; minute; team objective	Weight attack vs defence contextually
Risk/availability	Fitness flag; card state; defensive cost; eligibility	Disqualify unavailable; flag tradeoffs
Recommendation confidence	Data coverage; role similarity; match-specific sample size	Low/medium/high heuristic, NOT statistical probability

Recommended option fields: option_id, match_id, asof_sec, outgoing_player_id, incoming_player_id (nullable for KEEP), eligibility_status, attack_delta_estimate, control_delta_estimate, defensive_risk_delta, tactical_fit_score, estimated_confidence_band, supporting_event_ids, pros, cons, model_version. Estimated deltas are hypothetical modelling outputs; do not present them as validated causal effects.

8 | AI agent architecture and experience

Agent tools (read-only for MVP)

Tool	Input	Output
get_match_state	match_id, asof_sec	Score, formations, active players, available bench
get_player_performance	match_id, player_id, asof_sec	Current metrics and rating + source events
get_player_history	player_id, role, lookback	Comparable historical baseline + sample size
get_matchup_analysis	player_id, defender_id, match_id	Observed simulated duel data + historical context
evaluate_substitutions	match_id, team_id, asof_sec	Eligible keep/sub options, scores, risks, evidence
get_insight_candidates	match_id, since_sec	Statistical triggers and confidence flags

Agent loop: listen to event batch -> request deterministic state/feature tools -> determine novelty and importance -> seek supporting evidence -> generate structured insight payload -> run grounding and claims check -> adapt to fan/analyst tone -> publish. Do not allow the LLM to alter match facts or ratings.

Structured AI output contract


```
{
  "insight_id": "INS-067-001",
  "kind": "substitution_opportunity",
  "match_id": "SIM-001",
  "asof_sec": 4020,
  "audience": "fan",
  "headline": "A change on the right could help",
  "explanation": "The winger has completed 1 of 6 take-ons...",
  "evidence_event_ids": ["E-101", "E-117"],
  "recommendation_id": "REC-067-01",
  "uncertainty": "Limited matchup sample; tactical scenario",
  "generator_version": "agent-v1"
}
```


Audience surfaces

Fan: easy-to-understand explanations, what changed, notable moments, player stories. Analyst: rating breakdowns, form deviation, relevant event timeline, tactical matchup, substitution option ranking. Future broadcaster: timed captions, talking points and recap scripts with source-backed numbers.

9 | Synthetic generator and test strategy

Simulation and generation pipeline

Define 4 clubs and 100 fictional players -> sample attributes and role-specific priors -> schedule historical fixtures -> generate lineups and tactics -> simulate possession/transition state machine -> emit linked event sequences with consistent outcomes -> derive scores/statistics/ratings -> repeat with fixed seeds -> withhold demo/test scenario -> evaluate recommendations against baselines.

Validation checks - minimum required

Category	Example assertion
Relational integrity	Every event FK points to a known fixture/player/team; player belongs to a participating roster
Chronology	event_sec and sequence_no are ordered; no post-final events or negative minutes
Presence & eligibility	Only on-pitch players perform match actions; substitutions obey roster and window constraints
Possessions	Possession team changes on legitimate dispossessions, interceptions or restarts
Scoring consistency	Goals = scoreboard totals after excluding invalidated goals; assists must link to eligible goal
Stats reconciliation	Player and team shots, passes and cards match the underlying event ledger
Spatial and derived	Pitch coords in bounds; attacks oriented consistently; xG / xT recomputable
Grounding	Narrative claims reference observed or derived data; uncertainty for predictions
Fairness & robustness	Minimum-minutes gate, positional calibration, sparse-history fallbacks, no future leakage
Reproducibility	Same fixture seed + version reproduces same events and metric results

10 | Suggested implementation sequence

Phase	What to complete	Definition of done
A - Foundation	Schemas, teams/players/matches, generator seed, event validator	One fully reconciled synthetic match
B - Deterministic metrics	Aggregations, role rating weights, 10-match baselines	Reproducible ratings and comparisons
C - Intelligence	Rule-based insight triggers, matchup metrics, bench option evaluation	At least one credible no-change vs sub explanation
D - AI & UX	Foundry agent/tool calling, fan and analyst dashboards	Replay shows source-grounded, timed insights
E - Demonstration	Test scenarios and scripted 3-5 minute walkthrough	Clear problem, insight, recommendation and caveats

Division of work: BA/Data Analyst owns taxonomy, formulas, schema validation, quality scenarios, user stories and acceptance tests. Software Engineer owns simulator, persistence/API, Foundry tools, frontend, orchestrated replay and deployment. Both jointly approve the evidence and rating rules.

11 | Example relational records

players

```
player_id | team_id | display_name  | primary_position | archetype
P-ARS-07  | T-ARS   | Alex Mercer   | RW               | direct_winger
P-ARS-17  | T-ARS   | Theo Vale     | RW               | ball_carrier
P-MCI-03  | T-MCI   | Rowan Price   | LB               | defensive_fullback
```

events

event_id | match_id | sec | actor | type | outcome | defender
E-101 | SIM-001 | 3990 | P-ARS-07 | take_on | failed | P-MCI-03
E-117 | SIM-001 | 4004 | P-ARS-07 | take_on | failed | P-MCI-03

recommendation output

option | outgoing | incoming | likely upside | principal risk
KEEP | null | null | defensive continuity| low ball progression
SUB-OPTION-1 | P-ARS-07 | P-ARS-17 | direct ball carrying| defensive coverage

Acceptance criteria for the demo

- At least one scripted momentum or tactical alert is derived from event records and surfaced with evidence.
- At least one player-rating update is reproducible from source actions; baseline uses only prior matches.
- At least one bench comparison contains a keep option, eligible substitutions and clear pros/cons.
- Fan and analyst variants describe the same underlying evidence without contradicting it.
- The app explicitly labels the match and AI projections as synthetic; advice includes uncertainty.

Open decisions before detailed build

Confirm: fiction-only versus licensed likenesses; exact event definitions and coordinate convention; rating initial base and role weights; benchmark scenarios and success thresholds; substitute eligibility rules; xG/xT calculation or deferred status; intended deployment and storage stack. These are product decisions, not blockers to building the schema.